

Mortality Analysis and Gompertz Modeling Using R

Executive Summary

This project analyzes mortality experience using a life table containing mortality rates, survivor counts, and life expectancy values for male and female populations. The analysis was conducted in R using statistical and visualization techniques commonly employed in actuarial science.

Several key actuarial metrics were calculated, including remaining life expectancy at retirement age, survival probabilities, and median survival ages. In addition, a Gompertz mortality model was fit to adult mortality data to evaluate how closely observed mortality follows the expected exponential increase with age.

The analysis found that females experience lower mortality rates and greater longevity than males throughout most of the lifespan. Females were also more likely to survive to retirement age and exhibited a higher median survival age. The Gompertz model provided an excellent fit to observed adult mortality experience, supporting the well-established actuarial assumption that mortality increases approximately exponentially with age.

Overview

Mortality analysis is one of the foundational areas of actuarial science. Life tables are used extensively in insurance pricing, pension valuation, retirement planning, and risk management. These tables summarize mortality experience by providing probabilities of death, survivor counts, and remaining life expectancy for individuals at different ages.

The objective of this project was to analyze mortality patterns using a published life table and to investigate differences between male and female mortality experience. A secondary objective was to fit a Gompertz mortality model to adult mortality data and evaluate its ability to describe observed mortality rates.

The project was implemented entirely in R using the ggplot2 visualization package and standard statistical modeling techniques.

Data

The dataset consists of a life table containing ages 0 through 119 and includes the following variables:

- Age
- Male probability of death (q_x)
- Male survivor count (l_x)
- Male remaining life expectancy (e_x)
- Female probability of death (q_x)
- Female survivor count (l_x)
- Female remaining life expectancy (e_x)

The data was retrieved from the Social Security Administration Website. The data with a hypothetical cohort of 100,000 lives and tracks mortality experience across the lifespan. Survivor counts decrease as age increases, while mortality rates generally increase. Before the analysis was performed, the data were imported into R and converted into numerical format where necessary.

Methodology

I performed several analyses using the 2022 period life table for the Social Security area population.

First, mortality curves were generated to visualize age-specific probabilities of death. Separate plots were created for male mortality and for comparisons between male and female mortality experience.

Second, survival functions were calculated using the life table survivor counts. Survival probabilities were obtained by dividing each survivor count by the initial cohort size of 100,000.

Third, remaining life expectancy was examined across the lifespan. Special attention was given to life expectancy at age 65 because of its importance in retirement and pension applications.

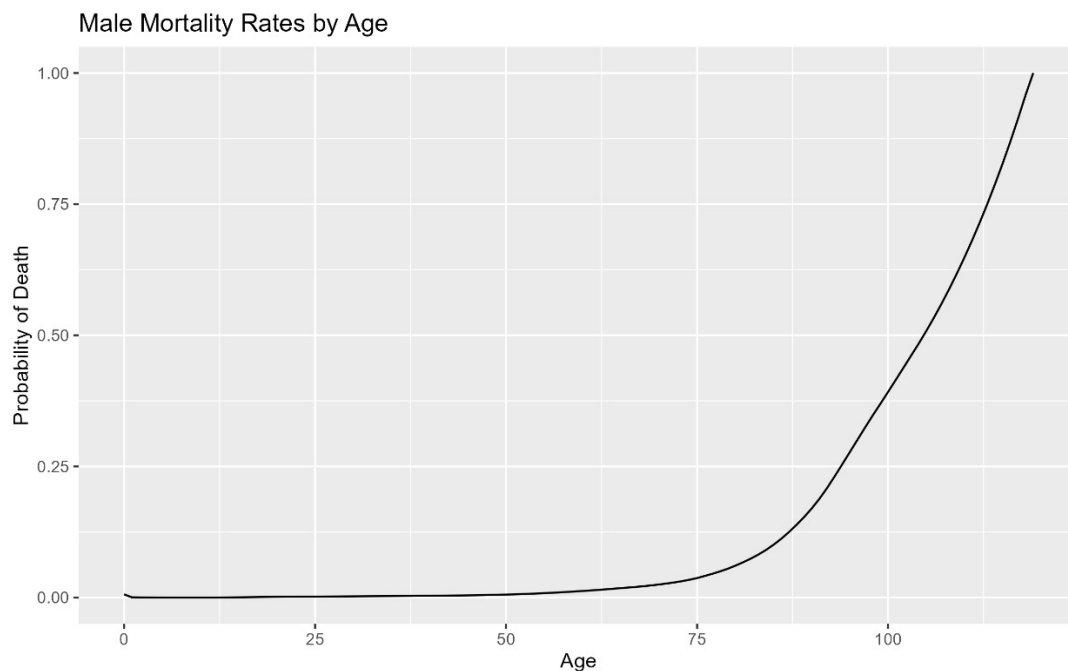
Finally, a Gompertz mortality model was fit to adult mortality data between ages 30 and 90. The Gompertz model assumes that mortality increases exponentially with age and is widely used in actuarial and demographic studies.

Results

Mortality Experience

Figure 1 displays male mortality rates by age.

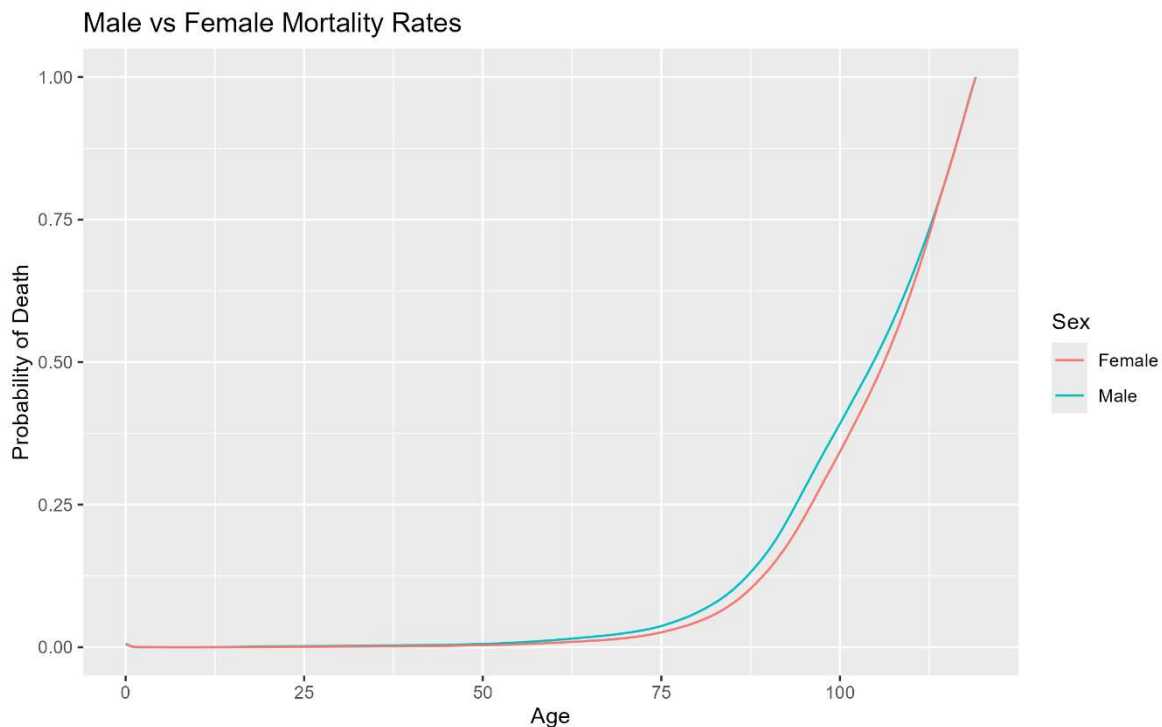
Mortality is relatively high during infancy, declines rapidly during childhood, and remains low throughout early adulthood. Beginning in middle age, mortality rates increase steadily and accelerate at older ages.



Male and Female Mortality Comparison

Figure 2 compares male and female mortality rates across the lifespan.

Female mortality rates remain below male mortality rates for most ages, indicating a longevity advantage. This pattern is commonly observed across developed populations and contributes to higher female life expectancy.



Survival Analysis

Figure 3 illustrates the male survival function, while Figure 4 compares male and female survival probabilities.

The analysis found:

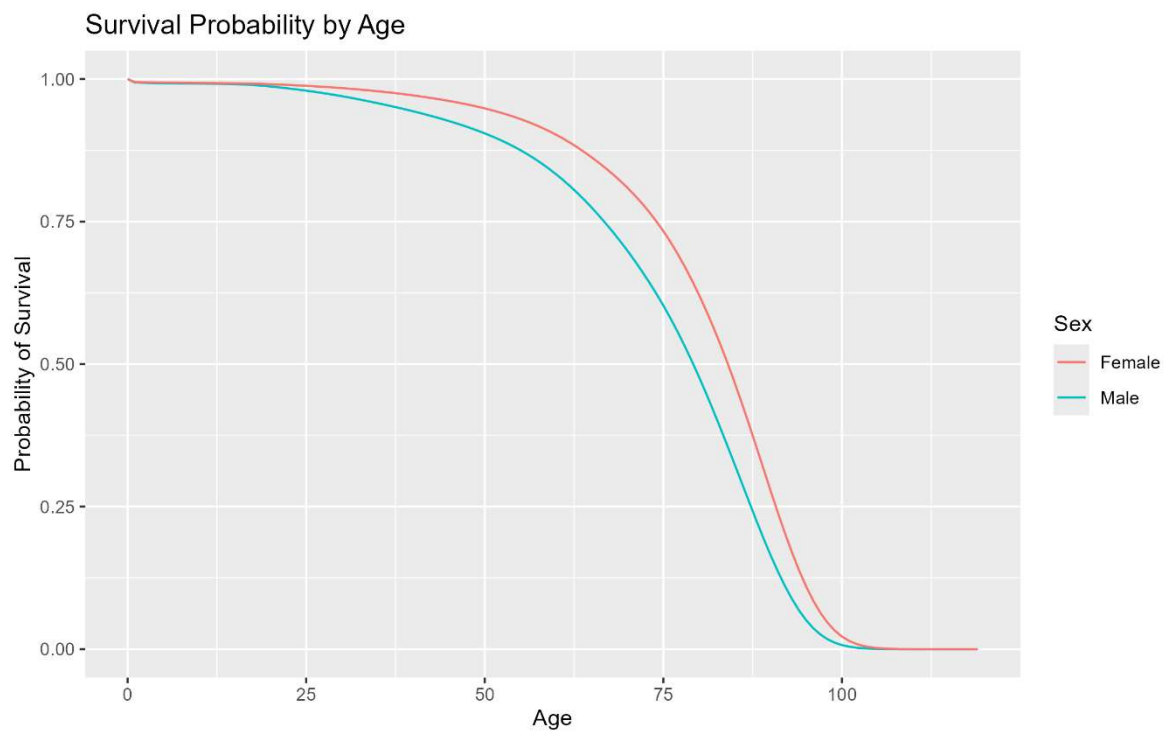
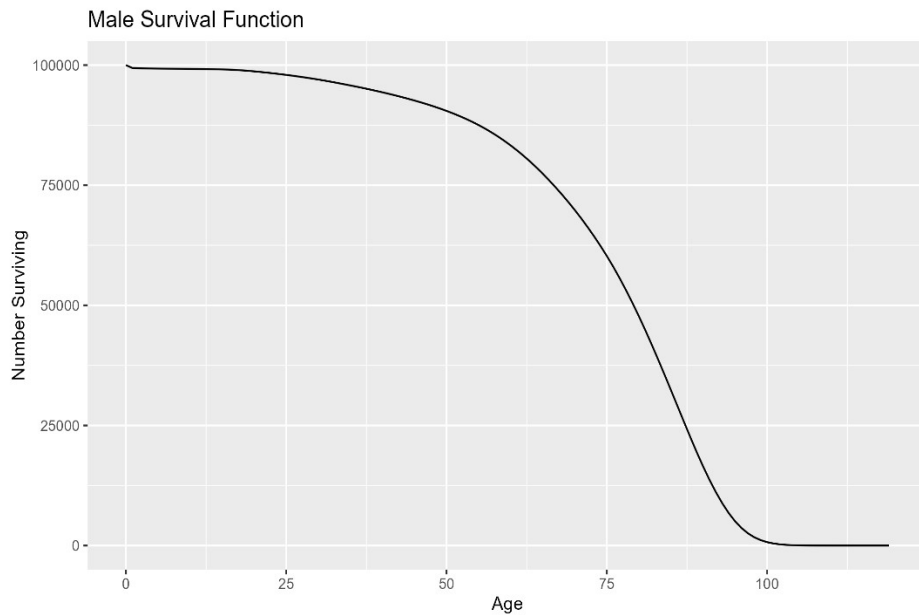
- Probability of surviving to age 65 (Male): 77.4%
- Probability of surviving to age 65 (Female): 86.2%

The difference demonstrates that females have a substantially greater likelihood of reaching retirement age.

Median survival ages were also calculated:

- Male median survival age: 80
- Female median survival age: 85

These results indicate that half of all males survive beyond age 80, while half of all females survive beyond age 85.



Life Expectancy Analysis

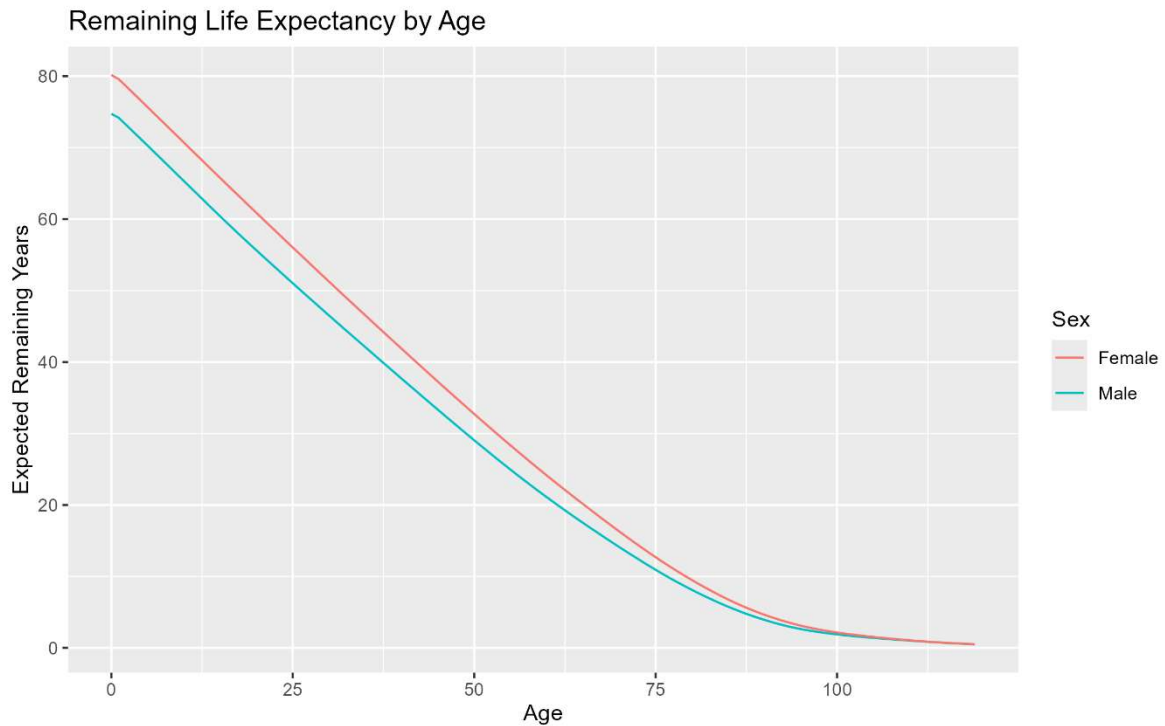
Remaining life expectancy was examined across ages.

At age 65:

- Male remaining life expectancy: 17.48 years
- Female remaining life expectancy: 20.12 years

The difference of 2.64 years highlights the persistent longevity advantage observed among females.

Figure 5 shows remaining life expectancy across the lifespan for both sexes.



Gompertz Mortality Model

A Gompertz model was fit using adult male mortality data from between the ages of 30 and 90.

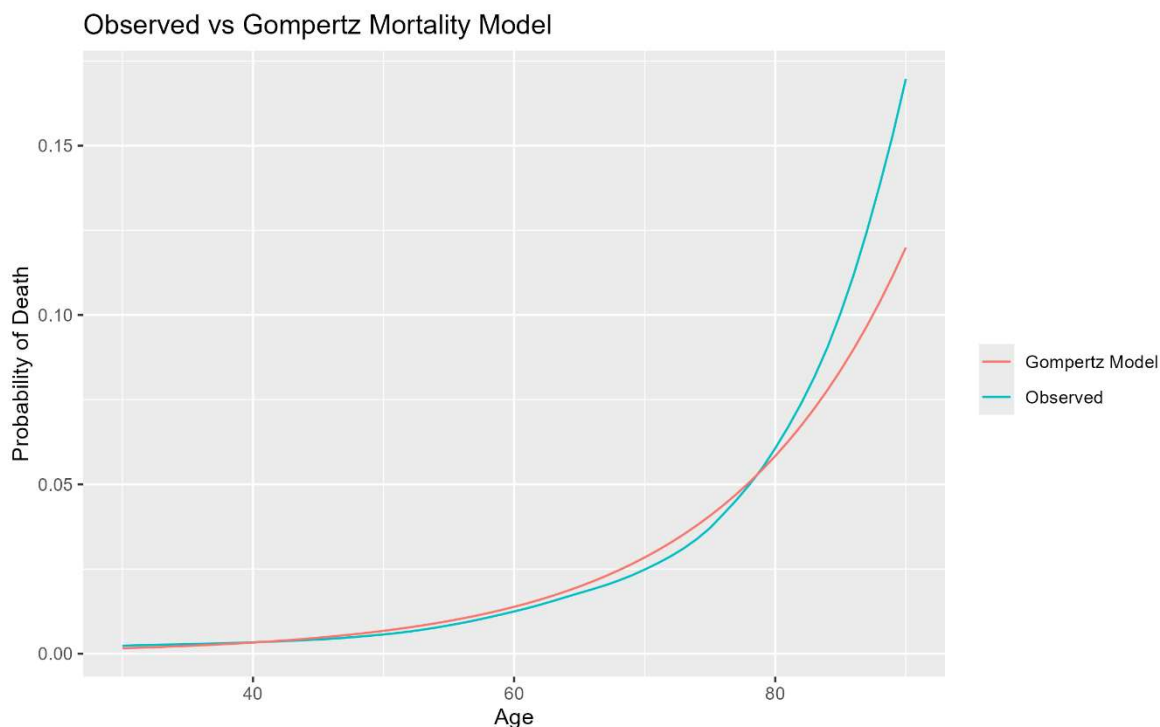
The estimated model was:

$$\log(qx) = -8.602 + 0.072x$$

The model achieved an R-squared value of 0.983 which indicates that approximately 98.3% of the variation in log mortality rates is explained by age only, which suggests an exceptionally strong fit.

Figure 6 compares observed mortality rates with those predicted by the Gompertz model.

The close agreement between observed and predicted values supports the actuarial assumption that mortality increases approximately exponentially throughout adulthood.



Discussion

Several important conclusions emerge from the analysis.

First, mortality follows the expected life-cycle pattern characterized by elevated infant mortality, low mortality throughout youth and early adulthood, and accelerating mortality at older ages.

Second, females consistently outperform males across multiple longevity measures. Females exhibit lower mortality rates, higher survival probabilities, longer remaining life expectancy, and a higher median survival age.

Third, the Gompertz model successfully captures the relationship between age and mortality. The high R-squared value demonstrates that age alone explains most of the variation in adult mortality rates within the observed range.

These findings align with established actuarial and demographic theory and demonstrate the usefulness of life tables in understanding population mortality dynamics.

Conclusion

This project applied actuarial and statistical techniques to analyze mortality experience using life table data. Mortality patterns, survival probabilities, and life expectancy measures were examined for both male and female populations.

Overall, the analysis found meaningful longevity advantages for females, including higher survival probabilities and greater remaining life expectancy. A Gompertz mortality model was successfully fit to adult mortality data and achieved an R-squared value of 0.983, indicating an excellent fit.